Homework 1

Module 1 · Homework · No date

Name _____ Date _____

1. A pharmacist dissolved 200 milligrams of a new antibiotic drug into a volume of purified water and placed the solution in a refrigerator (4°C). At various time intervals, the pharmacist removed a small aliquot from the solution and measured the amount of drug contained in each aliquot. The following data were obtained.

Time (hr)	Drug (mcg/mL)
0.5	233.3
1	217.6
2	189.5
4	143.6
8	82.5
12	47.4
16	27.2

1a Module 1

Is the decomposition of this antibiotic a first-order or a zero-order process?

A. Neither: it needs a third plot

B. Zero order

C. First order

1b Module 1

What is the rate constant k for the decomposition of this antibiotic?

1c Module 1

What is the half-life $t_{1/2}$?

1d Module 1

What was the initial volume of water used to prepare the original solution?

1e Module 1

How much time is required for the concentration of drug to decrease by 75%?

2. Below is the decrease in the amount of drug as a function of time.

Time (hr)	Drug A (mg)
0.5	489
1	478
2	455
3	433
6	365
9	298
12	230

Time (hr)	Drug A (mg)
16	140

2a Module 1

Does the decrease in the amount of drug A appear to be a zero-order or a first-order process?

A. First order

B. Zero order

C. Cannot tell without logarithms

2b Module 1

What is the rate constant k?

2c Module 1

What is the half-life $t_{1/2}$?

3. A solution of a drug was freshly prepared at a concentration of 315 mg/mL. After 30 days at 25°C, the drug concentration in the solution was 75 mg/mL.

3a Module 1

Assuming first-order kinetics, when will the drug decline to one-half of the original concentration?

3b Module 1

Assuming zero-order kinetics, when will the drug decline to one-half of the original concentration?